EKF Localization Filters

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***Abstract*— This report presents the implementation and analysis of an Extended Kalman Filter (EKF) for robot localization using a BNO055 IMU and Lidar sensor data. Data file 2020_2_26__17_21_59_filtered.csv was collected from a robot navigating a 10×10 meter square path with a pylon landmark at position (5.0m, −5.0m). The EKF successfully tracked the robot's position with mean tracking errors of 0.181 m when corrected every step, and 0.176m when corrected every other step, demonstrating that localization accuracy remains robust even with a 50% reduction in measurement updates.**

I. Introduction

This laboratory implements an Extended Kalman Filter (EKF) for robot localization, combining Inertial Measurement Unit (IMU) and Light Detection and Ranging (Lidar) sensor data. The robot travelled along a 10x10 meter square path in an open field, with a pylon landmark serving as the sole reference point for absolute position measurements.

The EKF algorithm operates in two steps: prediction using IMU acceleration and yaw velocity, and correction using transformed Lidar measurements. Two filtering strategies were compared: running corrections at every time step versus every other time step. This comparison reveals the trade-off between localization confidence and computational efficiency.

II. Coordinate systems and transformation

The global coordinate system is defined with the east direction as the positive x-axis, the north direction as the positive y-axis, yaw angle theta = 0º pointing east, and positive yaw angles measured counter-clockwise (CCW). The robot-centric coordinate system has forward direction (robot heading) as the local x-axis, left direction as the local y-axis, and IMU measures yaw with positive angles clockwise (CW). The Lidar outputs measurements in a frame where forward (relative to the robot) is the y-axis, and right (relative to the robot) is the x-axis.

III. Data Preprocessing

The filtered CSV contains 12 columns: $[X_{Lidar}, Y_{Lidar}, Z_{Lidar}, Timestamp, Latitude, Longitude, Yaw, Pitch, Roll, Accel_X, Accel_Y, Accel_Z]$ where Lidar measurements are in the robot frame, and accelerations represent forward (X) and left (Y) directions. The IMU yaw measurements required conversion from the clockwise positive convention to the global frame's counter-clockwise positive convention:

$$\theta_{global} = -\theta_{IMU} \quad [1]$$

Additionally, angles were converted from degrees to radians. Raw IMU acceleration data was filtered using a 9-sample moving average filter to reduce sensor noise in the acceleration measurements:

$$a_{filt,i} = \frac{1}{w} \sum_{j=i-w+1}^{i} a_{raw,j} \quad [2]$$

This smoothing improved velocity and position estimates by reducing high-frequency noise while preserving the underlying motion dynamics. The yaw rate ω was computed by finite difference of consecutive yaw measurements:

$$\omega[i] = \frac{\theta[i] - \theta[i-1]}{\delta t} \quad [3]$$

Where δt = 0.1 seconds (10Hz sampling rate). The wrap_to_pi function ensured angle differences remained in [-π , π]:

$$\text{wrap_to_pi}(\theta) = ((\theta + \pi)\ mod\ 2\pi) - \pi \quad [4]$$

Lidar detects the pylon in the robot's local frame. These measurements were transformed into direct observations of the robot's global position using the known landmark location (5.0, -5.0):

$$z_{x,t} = 5 - [y_{lidar} cos(\theta_t) + x_{lidar} sin(\theta_t)] \quad [5]$$

$$z_{y,t} = -5 - [y_{lidar} sin(\theta_t) - x_{lidar} cos(\theta_t)] \quad [6]$$

$$z_{\theta,t} = \theta_{lidar,t} \quad [7]$$

These equations convert the position of the landmark relative to the robot to the robot's position relative to the landmark.

The transformed measurements yielded:

$z_x range: [-1.078m, 10.698m]$ $(width: 11.78m)$

$z_y range: [-10.093m, 0.690m]$ $(width: 10.78m)$

The measurements successfully form a box shape with dimensions close to the expected 10x10 meter square.

IV. EKF Algorithm Design

A. State Vector

The state vector x has dimensions 5x1:

$$x = [x, y, v_x, v_y, \theta]^T \quad [8]$$

where x and y are the robot position in the global frame (meters), v_x and v_y are the robot velocity in the global frame (m/s), and θ is the robot yaw angle (radians, CCW positive).

B. Prediction Step - Motion Model

The prediction step uses a non-holonomic motion model with no lateral slip; the robot's movement is constrained to the direction it's facing as the wheels cannot slide sideways. Since the IMU measures acceleration in the robot's local frame, these must be rotated into the global frame:

$$a_{x,global} = a_{x,robot} cos(\theta) - a_{y,robot} sin(\theta) \quad [9]$$

$$a_{y,global} = a_{x,robot} sin(\theta) + a_{y,robot} cos(\theta) \quad [10]$$

For example, if the robot faces northeast (θ = 45º) and accelerates forward, this contributes equally to eastward and northward acceleration in the global frame. The state updates follows Newton's equations of motion: velocity changes due to acceleration, and position changes due to velocity:

$$v_{x,t} = v_{x,t-1} + a_{x,t} cos(\theta_{t-1})\delta t \quad [11]$$

$$v_{y,t} = v_{y,t-1} + a_{x,t} sin(\theta_{t-1})\delta t \quad [12]$$

$$x_t = x_{t-1} + v_{x,t-1}\delta t + 0.5 a_{x,global}\delta^2 t \quad [13]$$

$$y_t = y_{t-1} + v_{y,t-1}\delta t + 0.5 a_{y,global}\delta^2 t \quad [14]$$

$$\theta_t = \theta_{t-1} + \omega\delta t \quad [15]$$

Under the non-holonomic (no lateral slip) assumption, lateral acceleration $a_{y,robot} = 0$, so Equations 11-12 use only the forward component $a_{x,t}$ The motion model (Equations 9-15) is nonlinear due to the cos and sin terms. The Extended Kalman Filter handles this by using the full nonlinear equations to predict the state, and linearizing around the predicted state to update covariance.

The Jacobian matrix F is a 5x5 matrix of partial derivatives that captures how small changes in each state variable propagate to all other state variables.

$$F = \frac{\partial f}{\partial x} \quad [16]$$

Where f is the motion model function. The covariance prediction P propagates uncertainty forward:

$$P_{t|t-1} = FP_{t-1|t-1}F^T + Q \quad [17]$$

where FPF^T is old uncertainty propagating through motion, and Q is the new uncertainty from imperfect odometry and sensors. The process noise Q was tuned to:

- $\sigma^2_x = \sigma^2_y = 0.01m^2$ (position process noise)
- $\sigma^2_{vx} = \sigma^2_{vy} = 0.25\, m^2/s^2$ (velocity process noise)
- $\sigma^2_\theta = 0.01\, rad^2$ (heading process noise)

The Jacobian F captures how uncertainties in each state variable affect all other state variables. Q captures new uncertainty added by imperfect odometry.

C. Correction Step - Measurement Model

After the transformation (Equations 5-7), measurements directly observe x, y and θ. The measurement model is simply:

$$h(x) = [x, y, \theta]^T \quad [18]$$

This is linear, so the Jacobian H is straightforward:

$$H = \begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{vmatrix} \quad [19]$$

The Kalman gain K determines how much to trust the new measurement versus the prediction, and is computed as:

$$S = HP_{t|t-1}H^T + R \text{ (innovation covariance)} \quad [20]$$

$$K = P_{t|t-1}H^T S^{-1} \text{ (Kalman gain)} \quad [21]$$

Where R is the measurement noise covariance:

- $\sigma^2_{zx} = \sigma^2_{zy} = 0.09\, m^2$ (Lidar position noise)
- $\sigma^2_{z\theta} = 0.0025\, rad^2$ (IMU yaw noise)

The state and covariance updates are:

$$y = z - h(x_{t|t-1}) \quad \text{(innovation)} \qquad [22]$$

$$x_t = x_{t|t-1} + Ky \quad \text{(state update)} \qquad [23]$$

$$P_t = (I - KH)P_{t|t-1} \quad \text{(covariance update)} \qquad [24]$$

The innovation y represents how much the actual measurement differs from what we predicted. The Kalman gain K weights this: if measurements are noisy (large R), K is small and predictions are trusted more; if predictions are uncertain (large P), K is large and measurements are trusted more.

V. Results

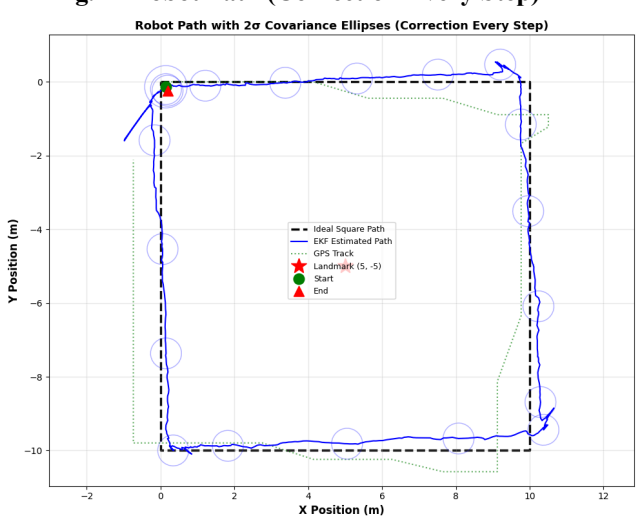
A. Correction Every Step

Running corrections at every time step achieved:

- Mean tracking error: 0.181m
- Maximum tracking error: 0.982 m
- RMS tracking error: 0.219m
- Final position (0.19m, -0.24m)
- Final position uncertainty: $\sigma_x = \sigma_y = 0.209m$

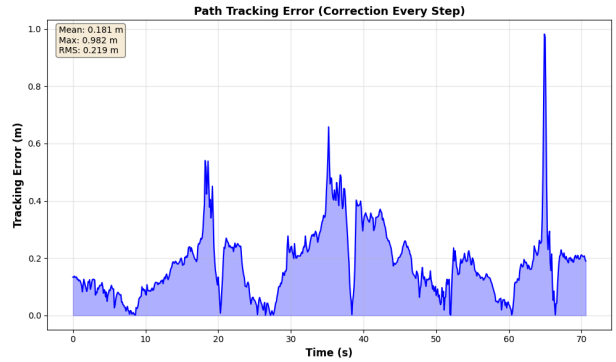
Figure 1 shows the estimated path closely following the ideal 10 x 10 meter square. The covariance ellipses indicate position uncertainty remaining relatively constant after an initial convergence period.

Fig. 1- Robot Path (Correction Every Step)



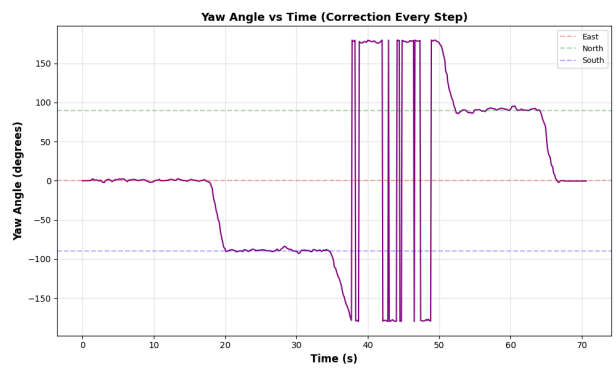
The tracking error in Figure 2 exhibits periodic spikes corresponding to corners of the square path, where the robot's dynamics change rapidly. On straight segments, the error remains below 0.3m. This pattern reflects the challenge of modeling sharp turns with a smooth motion model.

Fig. 2- Tracking Error (Correction Every Step)



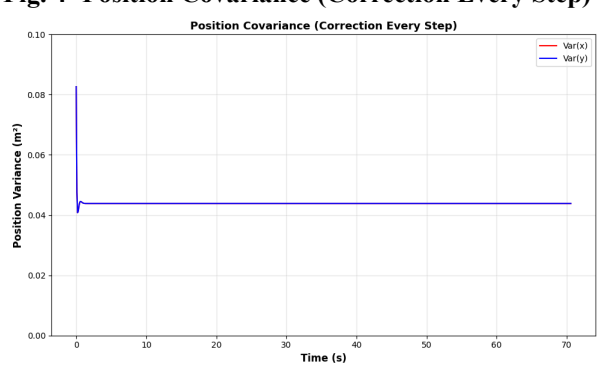
The yaw angle (Figure 3) shows clear transitions at 0º, 90º, 180º, and -90º, corresponding to the four cardinal directions as the robot traverses the square.

Fig.3- Yaw Angle (Correction Every Step)



Figures 4-6 show covariance evolution. Position covariance (Figure 4) converges rapidly from initial uncertainty and stabilizes around 0.044m².

Fig. 4- Position Covariance (Correction Every Step)



Velocity covariance (Figure 5) shows an initial transient before stabilizing at ~1.02m²/s². Yaw covariance (Figure 6) decreases to a steady value of ~ 0.002 rad².

Fig. 5- Velocity Covariance (Correction Every Step)

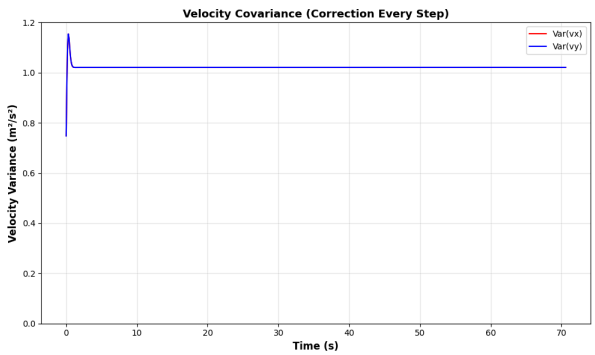
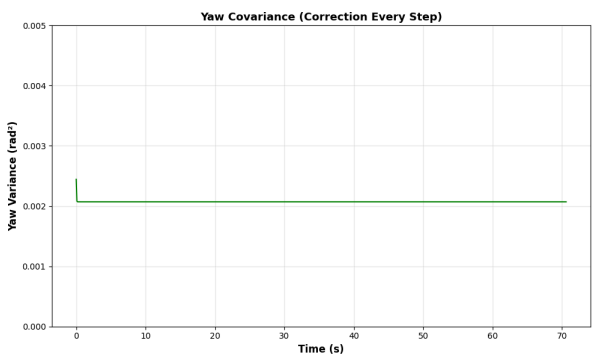


Fig. 6- Yaw Covariance (Correction Every Step)



B. Correction Every Other Step

Applying corrections only at alternative time steps (50% reduction) yielded:

- Mean tracking error: 0.176 m
- Maximum tracking error: 0.992 m
- RMS tracking error: 0.218 m
- Final position: (0.20m, -0.24m)
- Final position uncertainty: $\sigma_x = \sigma_y = 0.244m$

Figures 7-9 show results nearly identical to the full-rate correction case for path tracking and yaw estimation. However, the covariance plots (Figures 10-12) reveal important differences. Position covariance (Figure 10) grows significantly larger, reaching ~0.10m², compared to 0.044m² for full-rate corrections: a 2.3x increase. The y-axis scale in figure 10 compresses the sawtooth oscillation that is visible when zoomed in (see Figure 17). Velocity and yaw covariances similarly show increased uncertainty. Final yaw covariance (Figure 12) reaches ~ 0.012 rad², 6x larger than full-rate, though this increased uncertainty doesn't significantly degrade tracking accuracy.

Fig. 7- Robot Path (Correction Every Other Step)

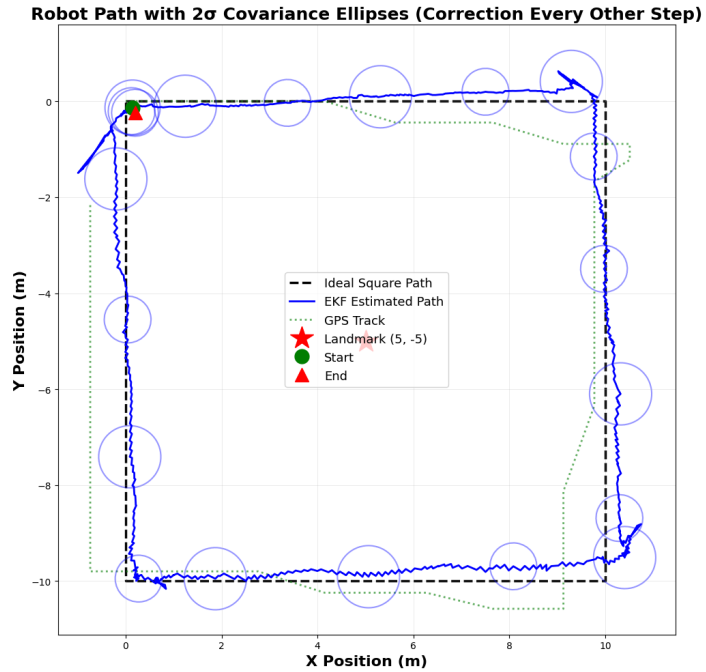


Fig. 8- Tracking Error (Correction Every Other Step)

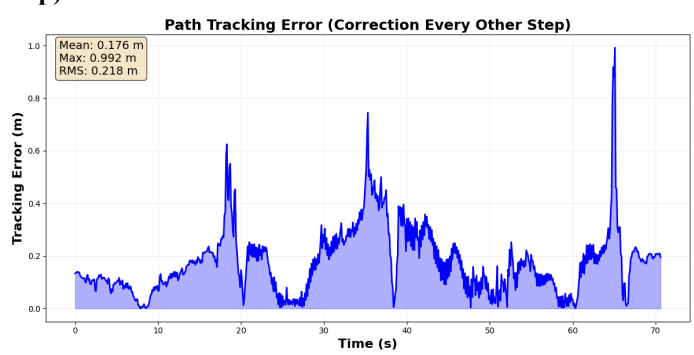


Fig.9- Yaw Angle (Correction Every Other Step)

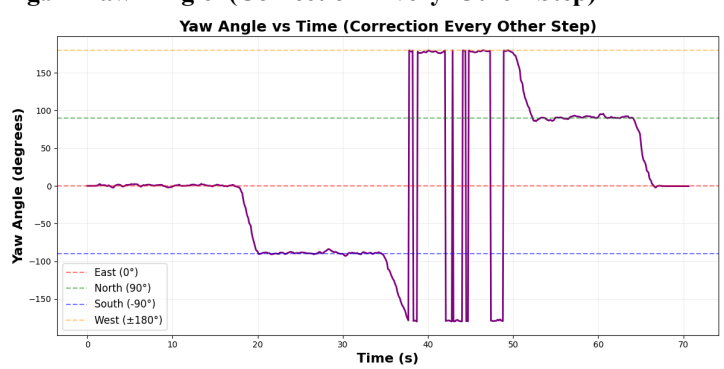


Fig. 10- Position Covariance (Correction Every Other Step)

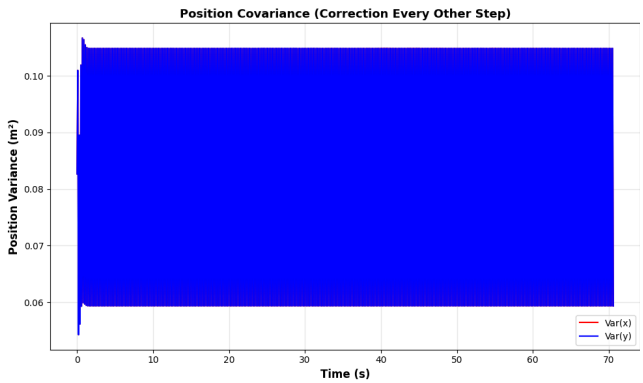


Fig. 11- Velocity Covariance (Correction Every Other Step)

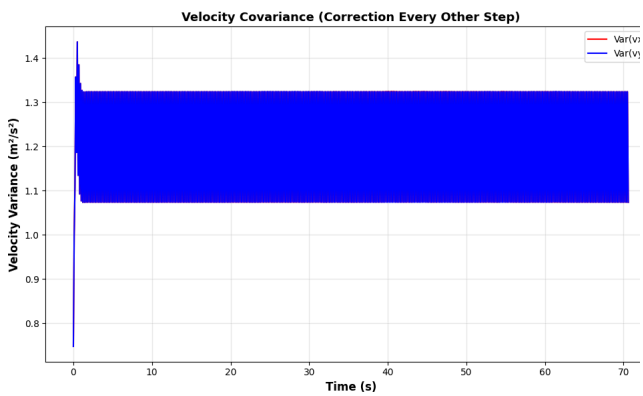
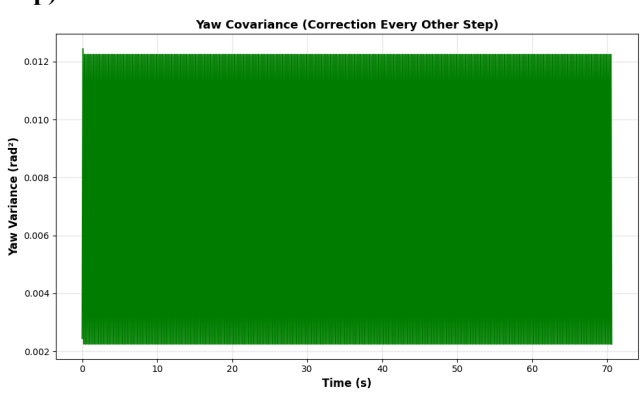


Fig. 12- Yaw Covariance (Correction Every Other Step)



C. Comparison

Figures 13-17 directly compare the accuracy of both correction frequencies utilized, with correction at every step displayed in blue, and correction at every other step displayed in red. Both paths are nearly indistinguishable and track the ideal square path closely.

Tracking error profiles (Figure 14) are also similar, with both methods displaying the same peaks at corners, and low error along straight segments. The every other step method shows a slightly higher error variance, but similar mean error. Both methods produce similar velocity estimate profiles (Figure 15), with the robot accelerating along straight paths and decelerating at corners. Peak velocity reaches 2.5 m/s.

Position uncertainty growth (Figure 16-17) shows key differences. The every step method maintains low uncertainty (blue, ~0.09m² trace). The every other step method shows substantial growth in uncertainty up to ~0.22m² (red trace). Figure 17 zooms into a 10-second window, revealing visible sawtooth oscillations as uncertainty grows during predictions (no measurements) and shrinks during corrections (measurements incorporated).

Fig. 13- Robot Path Comparison

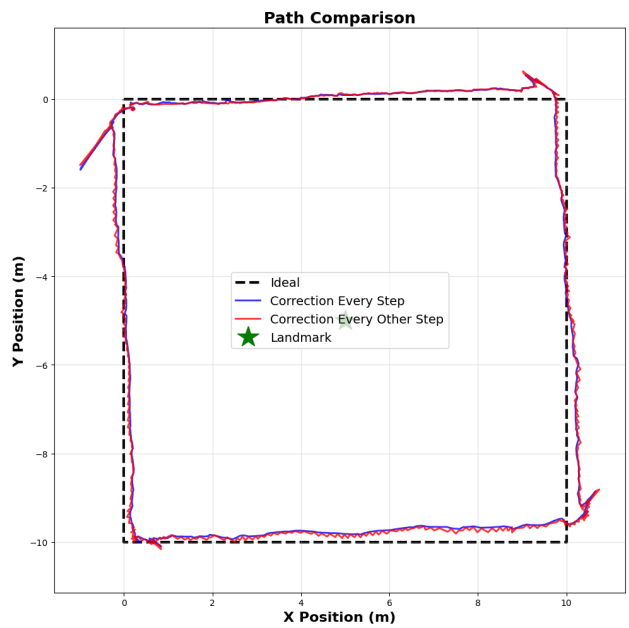


Fig. 14- Tracking Error Comparison

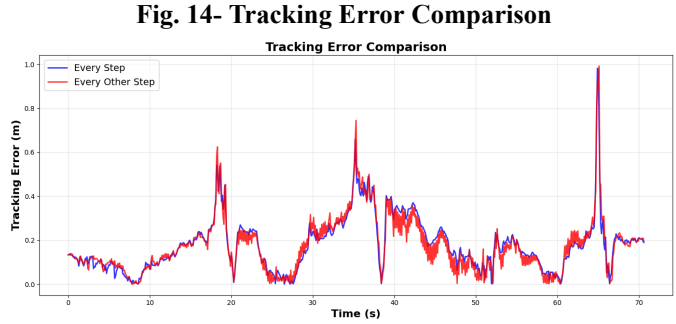


Fig. 15- Velocity Estimate Comparison

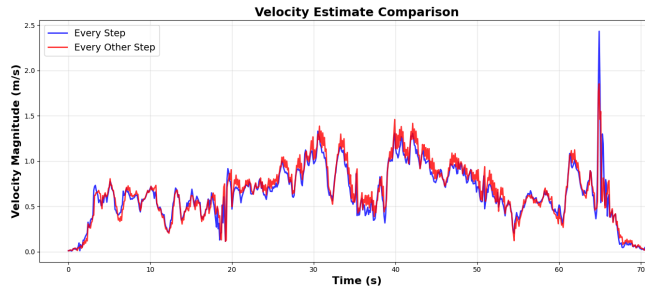


Fig. 16- Position Uncertainty Growth Comparison

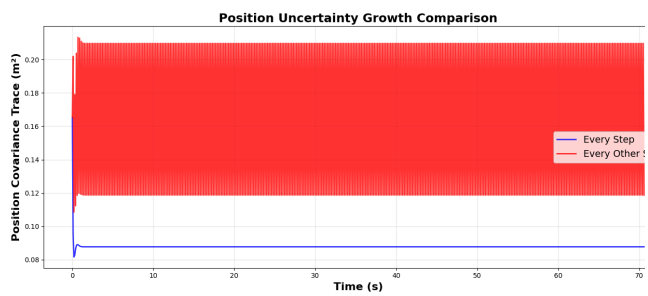
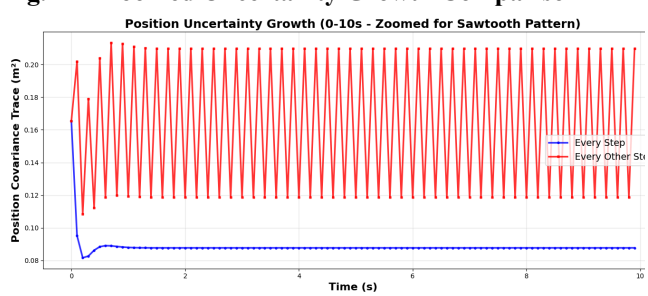


Fig. 17- Zoomed Uncertainty Growth Comparison



VI. Discussion

Despite halving the correction rate, tracking accuracy remained nearly unchanged (0.181m vs 0.176m mean error). This demonstrates that the 10Hz IMU provides sufficient accuracy for predictions over the 0.2-second intervals between Lidar updates. While path accuracy was maintained, position uncertainty increased by 135% (0.044 m² to 0.10 m²), illustrating that predictions increase uncertainty (through process noise Q), while corrections decrease it. The sawtooth pattern in covariance plots visually demonstrates this cycle. Reducing corrections to every other step halves the computational cost of matrix inversions and Kalman gain calculations. For resource-constrained systems, this 50% reduction could be significant while maintaining acceptable accuracy.

VII. Conclusion

This laboratory successfully implemented an Extended Kalman Filter for robot localization, achieving mean tracking errors of 0.18m while navigating a 10x10 meter square path. The EKF effectively fused IMU and Lidar data through motion and measurement models, with attention to coordinate transformations.

Comparisons of correction frequencies revealed that localization accuracy is robust despite 50% reductions in measurement updates, with tracking error remaining nearly constant. However, state uncertainty grows substantially (135% increase), highlighting the importance of regular corrections for maintaining confidence in state estimates. These results demonstrate key insights for robotics applications:

1) High-frequency IMU enables accurate short-term predictions between absolute measurements.
2) Periodic Lidar updates prevent long-term drift.
3) Correction frequency can be adapted based on whether computational efficiency or accuracy is prioritized.
4) The EKF framework quantifies and propagates uncertainty.

Future work could explore adaptive correction rates that increase during challenging maneuvers such as corners, and decrease during straight-line motion, optimizing the accuracy-efficiency tradeoff.

References

[1] Thrun, S., Burgard, W., & Fox, D. (2005). Probabilistic Robotics. MIT Press.

[2] Lab Notebook https://colab.research.google.com/drive/15mdntJiLFZ33NYKZu8IqCOZIR99eJ_LZ?usp=sharing

[3] E205 Lab 3 https://drive.google.com/file/d/18EuI6ln5KX9ivM4PA1975Ds0sibfZqie/view?usp=sharing .